

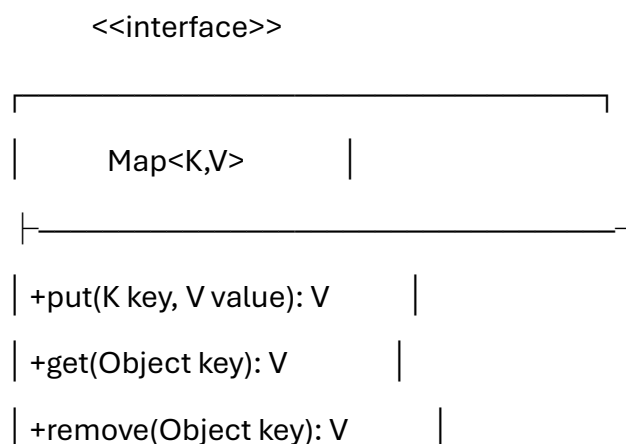
## Aim

To write a Java program that demonstrates the Map interface methods put(), get(), remove(), containsKey(), containsValue(), keySet(), values(), entrySet(), size(), isEmpty() and clear() using a HashMap.

## Algorithm

1. Start.
2. Import the java.util package.
3. Create a Map<Integer, String> object using HashMap, mapping student ID to student name.
4. Add key-value pairs using put() and display the map.
5. Call put() again with an existing key to replace its value. It returns the old value.
6. Retrieve values using get(). For a key that is not present it returns null.
7. Check for a key using containsKey() and for a value using containsValue().
8. Display the keys using keySet(), the values using values() and the key-value pairs using entrySet().
9. Traverse the entrySet() with a for-each loop, using getKey() and getValue() to print each pair.
10. Display the number of mappings using size() and check isEmpty().
11. Remove a mapping using remove(key), which returns the removed value, then display the map.
12. Remove all mappings using clear() and check size() and isEmpty() again.
13. Stop.

## UML Diagram



```

| +containsKey(Object key): boolean |
| +containsValue(Object value): boolean |
| +keySet(): Set<K> |
| +values(): Collection<V> |
| +entrySet(): Set<Map.Entry<K,V>> |
| +size(): int |
| +isEmpty(): boolean |
| +clear(): void |
|_____|

```

```

    ▲      ▲      ▲
    |      |      | (implements)
HashMap  LinkedHashMap  TreeMap

```

```

    ▲
    | (uses)
    |_____|
    | MapMethodsDemo |
    |_____|
    | +main(String[]) |
    |_____|

```

Map<K,V> contains nested interface <<interface>> Map.Entry<K,V>

```
+getKey(): K +getValue(): V
```

Code:

```
package Observations;
```

```
import java.util.*;
```

```
public class MapInterface {
```

```

public static void main(String[] args) {

    Map<Integer, String> map = new HashMap<>();

    // put()

    map.put(101, "Arun");
    map.put(102, "Bala");
    map.put(103, "Chitra");

    System.out.println("Map after put()    : " + map);


    // put() with an existing key replaces the value

    String old = map.put(102, "Balaji");

    System.out.println("put(102, \"Balaji\") returned: " + old);

    System.out.println("Map after replace  : " + map);


    // get()

    System.out.println("get(102)          : " + map.get(102));
    System.out.println("get(999)          : " + map.get(999));


    // containsKey() and containsValue()

    System.out.println("containsKey(101)   : " + map.containsKey(101));
    System.out.println("containsKey(999)   : " + map.containsKey(999));
    System.out.println("containsValue(\"Chitra\") : " + map.containsValue("Chitra"));
    System.out.println("containsValue(\"Dinesh\") : " + map.containsValue("Dinesh"));


    // keySet(), values(), entrySet()

    System.out.println("keySet()           : " + map.keySet());
    System.out.println("values()           : " + map.values());
    System.out.println("entrySet()          : " + map.entrySet());
}

```

```

// Traversing the entrySet

System.out.println("Traversing entrySet:");

for (Map.Entry<Integer, String> e : map.entrySet()) {

    System.out.println("Key: " + e.getKey() + ", Value: " + e.getValue());

}


// size() and isEmpty()

System.out.println("size()          : " + map.size());

System.out.println("isEmpty()       : " + map.isEmpty());


// remove()

System.out.println("remove(101)      : " + map.remove(101));

System.out.println("remove(999)      : " + map.remove(999));

System.out.println("Map after remove  : " + map);

System.out.println("size()          : " + map.size());


// clear()

map.clear();

System.out.println("Map after clear()   : " + map);

System.out.println("size()          : " + map.size());

System.out.println("isEmpty()       : " + map.isEmpty());

}

}

```

Output:

```

PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java> & 'C:\Program Files\Java\jdk-26.0.2.1\bin\
es' '-cp' 'C:\Users\Sahithi_pyla\AppData\Roaming\Code\User\workspaceStorage\234db3eb4e599b
bservations.MapInterface'
Map after put()           : {101=Arun, 102=Bala, 103=Chitra}
put(102, "Balaji") returned: Bala
Map after replace         : {101=Arun, 102=Balaji, 103=Chitra}
get(102)                  : Balaji
get(999)                   : null
containsKey(101)           : true
containsKey(999)           : false
containsValue("Chitra")    : true
containsValue("Dinesh")    : false
keySet()                   : [101, 102, 103]
values()                   : [Arun, Balaji, Chitra]
entrySet()                 : [101=Arun, 102=Balaji, 103=Chitra]
Traversing entrySet:
Key: 101, Value: Arun
Key: 102, Value: Balaji
Key: 103, Value: Chitra
size()                     : 3
isEmpty()                  : false
remove(101)                : Arun
remove(999)                : null
Map after remove           : {102=Balaji, 103=Chitra}
size()                     : 2
Map after clear()          : {}
size()                     : 0
isEmpty()                  : true
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>

```

Result:

Thus, the Java program demonstrating the Map interface methods was executed successfully and the output was verified.

- put() adds a new mapping, or replaces the value of an existing key and returns the old value.
- get() and remove() return null when the key is not present.
- keySet(), values() and entrySet() return views of the map's keys, values and key-value pairs.
- clear() removes all mappings, leaving the map with size() 0 and isEmpty() true.

HashMap does not guarantee any order, so on your system the order of entries in the output could differ. With small integer keys like these it normally matches the output above. If you need a fixed order, use LinkedHashMap (insertion order) or TreeMap (sorted by key).

